Volatility is rough: replication and forecasting

A 31-index replication of Gatheral, Jaisson & Rosenbaum (2018)
and an out-of-sample forecasting competition

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October 2026

Abstract

Using 5-minute realized variance for 31 equity indices (Oxford-Man Realized Library, 2000–2022) we re-estimate the roughness of log-volatility. The scaling $m(q, \Delta) \propto \Delta^{qH}$ holds well, and the median Hurst exponent is $H = 0.126$ (range 0.079–0.171; S&P 500: 0.149 ± 0.021), in line with the paper. Range-based (Garman–Klass) variance from daily OHLC, extended to October 2026 for 20 indices, gives a lower median $H = 0.077$, consistent with measurement noise pulling H down. In a walk-forward competition (validation up to 2014, untouched test period from 2015), the paper's RFSV predictor is *not* better than HAR at 1 day (worse on 17 of 31 indices by Diebold–Mariano at 5%), is level with HAR at 5 days and marginally better on average at 20 days (not significant on any index). A HAR model with a leverage term and the RFSV forecast as an extra regressor, chosen on the validation period, lowers the test log-MSE relative to HAR by 5.9% (1 day) and, with a forecast combination, by 1.8% (5 days) on the RV data; it beats both HAR and RFSV on every index at 1 day. At 20 days the validation-chosen model (rolling-H RFSV) does not beat either benchmark on the test period.

1 Introduction

Gatheral, Jaisson and Rosenbaum (GJR, 2018) showed that the log of realized volatility behaves like a fractional Brownian motion (fBm) with Hurst exponent $H \approx 0.1$, far below the $H > 1/2$ of the long-memory models used before. Two consequences follow: volatility paths are *rough*, and the conditional expectation of future log-volatility has a closed form, which the authors use as a forecasting rule (the RFSV predictor) and find to be competitive with the HAR model of Corsi (2009). This project (i) replicates the estimation on all 31 indices of the Oxford-Man library and on a range-based series to 2026, (ii) checks two side claims (Gaussian increments and the smoothing bias), and (iii) runs a walk-forward forecasting competition with an honest validation/test split, including attempts to improve on both RFSV and HAR.

2 Data

Realized variance (RV). 5-minute realized variance (`rv5`) and open-to-close returns for 31 indices from the final public version of the Oxford-Man Institute Realized Library (Heber, Lunde, Shephard & Sheppard, 2009), 3 January 2000 – 25 February 2022, obtained from the Internet Archive copy of the official zip (snapshot of 1 March 2022). The library was discontinued in 2022 and has no open licence; it was free for academic research with citation, so the data are not redistributed with the code. **Range-based variance (GK).** Daily OHLC for 20 indices from Yahoo Finance (via `yfinance`), January 2000 – 12 October 2026. Daily variance is the Garman–Klass estimator $\frac{1}{2}(\log H/L)^2 - (2\log 2 - 1)(\log C/O)^2$; Parkinson's $(\log H/L)^2/(4\log 2)$ is a cross-check (correlation of logs 0.91–0.97). Days with zero range, prices outside $[L, H]$ or non-positive GK are dropped (at most 11 days per index). The correlation between logRV and logGK on overlapping days is 0.77–0.91.

The two data types are *never mixed* in one series; every result is reported separately for RV and GK. Throughout, $\log \sigma_t = \frac{1}{2} \log v_t$ with v_t the daily variance proxy, and time is counted in trading days of each index.

3 Replication

3.1 Scaling of log-volatility increments

For $q \in \{0.5, 1, 1.5, 2, 3\}$ and $\Delta = 1, \ldots, 50$ we compute $m(q, \Delta) = \frac{1}{N-\Delta} \sum_t |\log \sigma_{t+\Delta} - \log \sigma_t|^q$. If $\log \sigma_t - \log \sigma_s \sim \nu\, |t - s|^H Z$ with $Z \sim N(0, 1)$, then $m(q, \Delta) = \nu^q\, \mathbb{E}|Z|^q\, \Delta^{qH}$, so an OLS regression of $\log m(q, \Delta)$ on $\log \Delta$ has slope $\zeta_q = qH$. We estimate H as the slope of ζ_q on q through the origin, and ν from the $q = 2$ intercept. Standard errors come from a block bootstrap: the series is cut into 250-day blocks, the sums $\sum |\text{increment}|^q$ and pair counts are computed inside each block, and blocks are resampled with replacement (500 draws). Figure 1 shows the S&P 500 case: the $\log m$–$\log \Delta$ lines are straight and ζ_q is linear in q (monofractal), as in the paper.

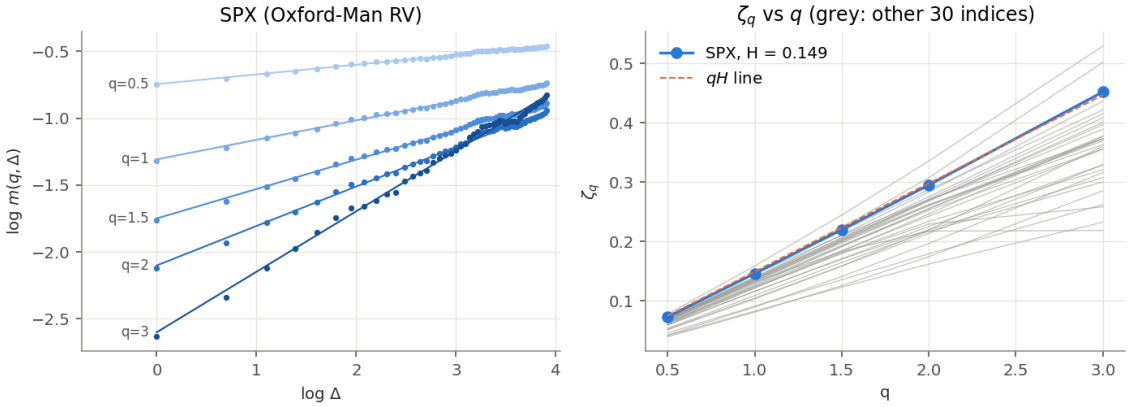


Figure 1: Left: $\log m(q, \Delta)$ against $\log \Delta$ for the S&P 500 (RV), with OLS fits. Right: ζ_q against q; SPX in blue, the other 30 indices in grey.

Index	H_{RV} (s.e.)	H_{GK} (s.e.)	ν_{RV}	Index	H_{RV} (s.e.)	H_{GK} (s.e.)	ν_{RV}
AEX	0.142 (0.019)	0.072 (0.013)	0.30	KSE	0.124 (0.017)	–	0.40
AORD	0.090 (0.021)	0.056 (0.015)	0.37	MXX	0.079 (0.011)	0.048 (0.008)	0.38
BFX	0.138 (0.017)	–	0.28	N225	0.134 (0.019)	0.075 (0.012)	0.32
BSESN	0.120 (0.010)	0.078 (0.011)	0.32	NSEI	0.107 (0.010)	0.072 (0.016)	0.35
BVLG	0.160 (0.046)	–	0.24	OMXC20	0.104 (0.013)	–	0.33
BVSP	0.133 (0.016)	0.067 (0.008)	0.31	OMXHPI	0.112 (0.017)	–	0.33
DJI	0.136 (0.020)	0.092 (0.015)	0.36	OMXSPI	0.127 (0.018)	–	0.32
FCHI	0.133 (0.017)	0.078 (0.011)	0.30	OSEAX	0.108 (0.020)	–	0.35
FTMIB	0.127 (0.022)	–	0.32	RUT	0.100 (0.022)	0.072 (0.011)	0.37
FTSE	0.111 (0.014)	0.075 (0.010)	0.35	SMSI	0.128 (0.013)	–	0.31
GDAXI	0.127 (0.018)	0.079 (0.011)	0.31	SPX	0.149 (0.021)	0.094 (0.014)	0.35
GSPTSE	0.114 (0.026)	0.079 (0.019)	0.37	SSEC	0.131 (0.010)	–	0.33
HSI	0.089 (0.013)	0.054 (0.007)	0.29	SSMI	0.171 (0.025)	0.086 (0.014)	0.25
IBEX	0.126 (0.019)	0.071 (0.012)	0.29	STI	0.101 (0.029)	–	0.25
IXIC	0.143 (0.019)	0.083 (0.011)	0.32	STOXX50E	0.093 (0.015)	0.083 (0.009)	0.38
KS11	0.123 (0.015)	0.078 (0.009)	0.28				
Median H: RV 0.126 (31 indices, range 0.079–0.171); GK 0.077 (20 indices, range 0.048–0.094).							

Table 1: Estimated H (block-bootstrap s.e.) from 5-minute RV (2000–2022) and from Garman–Klass variance (2000–2026), and the vol-of-log-vol ν (RV, daily units).

Table 1 reports every index. With RV, 25 of 31 estimates lie in $[0.10, 0.15]$ (4 below, 2 above), the median is 0.126 and none exceeds 0.18: the paper's finding is confirmed across markets. The bootstrap standard errors (typically 0.015–0.02) are larger than a naive i.i.d. calculation would give, because H moves between calm and stressed years. For a few indices (STOXX50E, RUT) ζ_q/q falls with q, i.e. the largest moves scale less steeply; for most it is flat. The GK estimates are systematically lower (median 0.077), and Figure 2 shows this holds index by index.

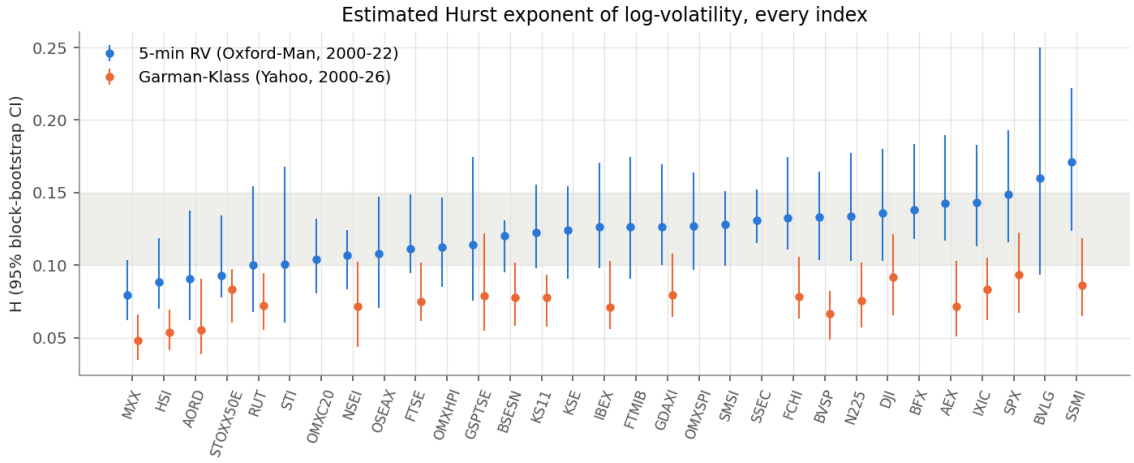


Figure 2: H with 95% block-bootstrap intervals for every index; the shaded band is $[0.10, 0.15]$.

3.2 Gaussianity of increments

GJR report that $\log \sigma_{t+\Delta} - \log \sigma_t$ is close to Gaussian at all lags. The histograms for SPX (notebook, Fig. `gauss_spx.png`) are bell-shaped and close to the normal density at $\Delta = 1, 5, 25, 125$. Table 2 gives the median skewness and excess kurtosis over indices: the skew is small and positive at every lag, but the RV increments have fatter tails than a Gaussian (excess kurtosis about 2 at one day, 0.8 at 125 days). The Gaussian description is therefore a good first approximation of the body, not of the tails. For GK the increments look *more* Gaussian, which is what one expects when a larger, roughly Gaussian measurement error is added.

	RV (31)		GK (20)	
Lag Δ	skew	exc. kurt.	skew	exc. kurt.
1	0.15	2.08	0.08	0.15
5	0.19	1.70	0.08	0.38
25	0.34	1.69	0.18	0.62
125	0.25	0.78	0.14	0.44

Table 2: Median skewness and excess kurtosis of log-vol increments.

H true	spot	integrated	integrated + noise
0.05	0.048	0.159	0.081
0.10	0.096	0.195	0.131
0.15	0.144	0.231	0.176
0.20	0.194	0.269	0.219
0.30	0.295	0.350	0.309
0.50	0.500	0.529	0.499

Table 3: Smoothing bias: mean $\hat{H}$ over 3 simulated RFSV paths ($\nu = 0.3$, 5000 days, 50 steps/day).

3.3 Smoothing bias

RV estimates the *integrated* variance $\int_{t-1}^{t} \sigma_s^2 ds$, not spot variance. We simulate $\log \sigma_t = \nu W_t^H$ exactly (Davies–Harte) on a 50-step intraday grid and estimate H from (i) the spot log-vol at day ends, (ii) the log of daily integrated variance, (iii) (ii) plus i.i.d. noise of s.d. 0.25 on log variance. Table 3: the estimator is accurate on spot volatility, daily averaging biases $\hat{H}$ *upwards* (0.10 becomes about 0.20 with these parameters), and measurement noise pulls it back down. The bias is largest for small H, so if anything the true H of spot volatility is smaller than the RV estimates, which

strengthens the roughness conclusion. The same mechanism explains why the noisier GK proxy gives lower $\hat{H}$ than RV: the two biases act in opposite directions and their balance depends on the proxy, so H from a daily proxy is not a pure property of the underlying volatility.

4 Forecasting method

4.1 The RFSV predictor

Let $X_t = \log \sigma_t$ and assume $X_t = X_0 + \nu W_t^H$ with W^H a fBm (the stationary RFSV model adds a very slow mean reversion that is negligible at horizons of days). For the Mandelbrot–Van Ness fBm $W_t^H = \frac{1}{C_H}\int_{-\infty}^{t}\big[(t-s)^{H-1/2} - (-s)_+^{H-1/2}\big]dW_s$, the filtration of W^H equals that of W. Splitting $W_{t+\Delta}^H$ into the part driven by dW_s, $s \le t$ (known at t) and the part driven by dW_s, $s \in (t, t+\Delta]$ (independent of $\mathcal{F}_t$, mean zero) gives, with $\mathbb{E}_t = \mathbb{E}[\cdot\,|\mathcal{F}_t]$ and $a_0 = C_H^{-1}\int_{-\infty}^{0}(-s)^{H-1/2}dW_s$,

$$\mathbb{E}_t[W_{t+\Delta}^H] = C_H^{-1}\int_{-\infty}^{t}(t+\Delta-s)^{H-1/2}dW_s - a_0, \tag{1}$$

$$\mathrm{Var}_t[W_{t+\Delta}^H] = \frac{1}{C_H^2}\int_0^{\Delta} u^{2H-1}du = c\,\Delta^{2H}. \tag{2}$$

Re-expressing the first integral in terms of the observed past of W^H instead of W (an inversion of the fractional integral; Nuzman & Poor, 2000) yields the formula used by GJR:

$$\mathbb{E}[\log v_{t+\Delta}\,|\,\mathcal{F}_t] = \frac{\cos(H\pi)}{\pi}\,\Delta^{H+1/2}\int_{-\infty}^{t}\frac{\log v_s}{(t-s+\Delta)(t-s)^{H+1/2}}\,ds. \tag{3}$$

The kernel integrates to one, so the forecast is a weighted average of past log variances with weights that decay like a power law: recent days dominate, but the whole past matters. With the normalisation $\mathrm{Var}(W_\Delta^H) = \Delta^{2H}$ one finds $c = \frac{1}{2HC_H^2} = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$ ($c = 1$ at $H = 1/2$; $c \approx 0.64$ at $H = 0.1$). Because $\log v = 2X$ is conditionally Gaussian with variance $4c\nu^2\Delta^{2H}$, the variance forecast is

$$\mathbb{E}[v_{t+\Delta}\,|\,\mathcal{F}_t] = \exp\Big(\mathbb{E}[\log v_{t+\Delta}|\mathcal{F}_t] + 2c\,\nu^2\Delta^{2H}\Big). \tag{4}$$

Implementation. The integral is truncated at $L = 500$ days. Cell k (lag $u \in [k, k+1)$) gets weight $w_k \propto \big[(k+1)^{1/2-H} - k^{1/2-H}\big]/\big[(1/2-H)(k+1/2+\Delta)\big]$, i.e. the singular factor $u^{-H-1/2}$ is integrated exactly; weights are normalised to sum to one. H and ν come from the $q = 2$ regression on the past only and are re-estimated every 20 days.

4.2 Benchmarks and protocol

RW: $\widehat{\log v}_{t+h} = \log v_t$. **HAR** (Corsi, 2009) on log variance, direct h-step: $\log v_{t+h} = \beta_0 + \beta_d \log v_t + \beta_w \overline{\log v}_{t-4:t} + \beta_m \overline{\log v}_{t-21:t} + \varepsilon$, re-fitted every day by OLS on all pairs whose target is already observed ($s + h \le t$); the variance forecast is $\exp(\hat{f} + \hat{s}^2/2)$. **GARCH(1,1)** (Gaussian QMLE, zero mean) on daily returns (open-to-close for RV, close-to-close for GK), re-fitted every 20 days; because return variance and RV differ in level, its log and level forecasts are calibrated by the in-sample mean of $\log v_s - \log \sigma_s^2$ and of v_s/σ_s^2 (past data only). **Split.** The first 1000 days of each index are a burn-in. *Validation*: forecasts whose target date is $\le$ 31-12-2014 (from December 2003). *Test*: origins from 1-1-2015 to the end of each sample (25-02-2022 for RV, October 2026 for GK). The target is the point value $\log v_{t+h}$, $h \in \{1, 5, 20\}$ as in GJR. **Losses**: MSE of the log-variance forecast (identical ratios for log-vol) and QLIKE $= v/\hat{v} - \log(v/\hat{v}) - 1$. Every model is scored on the same days. **Tests**: Diebold–Mariano with a Bartlett HAC variance, lag $\max(h, \lfloor 4(T/100)^{2/9} \rfloor)$. We report pairwise DM tests per index; a Model Confidence Set was not run, for cost and simplicity. **No look-ahead.** Every model receives only data up to the origin; an automated check perturbs every observation after a cut-off day and confirms that no forecast made at or before that day changes (it passes for all models and horizons).

5 Results

Table 4 gives, for the test period, the geometric mean over indices of each model's loss divided by HAR's loss (below 1 = better than HAR); Table 5 counts significant DM wins and losses.

	Oxford-Man RV (31 indices)						Garman–Klass (20 indices)					
	MSE / HAR			QLIKE / HAR			MSE / HAR			QLIKE / HAR		
Model	1	5	20	1	5	20	1	5	20	1	5	20
RW	1.338	1.391	1.444	1.458	1.534	1.907	1.499	1.564	1.600	2.146	2.348	2.649
GARCH	1.333	1.179	1.152	1.302	1.147	0.983	1.154	1.106	1.118	1.095	1.043	0.950
HAR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
RFSV	1.024	1.001	0.996	1.009	1.002	1.079	1.059	1.015	0.996	1.060	1.019	1.065
RFSV-roll250	1.031	1.014	**1.008**	1.015	1.024	1.144	1.062	1.021	1.001	1.089	1.029	1.101
RFSV-roll500	1.032	1.012	1.006	1.016	1.015	1.110	1.061	1.019	1.000	1.073	1.027	1.081
RFSV-roll1000	1.025	1.005	0.999	1.011	1.009	1.097	1.062	1.018	0.999	1.066	1.024	1.078
HAR-L	0.954	0.987	0.992	0.957	0.990	0.999	0.944	0.979	0.994	0.929	0.968	0.993
HAR-RFSV	0.990	0.989	0.981	0.993	1.009	1.036	0.989	0.988	0.979	0.993	1.010	1.023
HAR-L-RFSV	**0.941**	0.973	0.975	0.946	0.998	1.036	**0.933**	**0.965**	0.974	0.919	0.973	1.006
COMB(RFSV,HAR)	0.996	0.993	0.991	0.992	0.994	1.057	0.995	0.997	0.990	0.993	0.998	1.038
COMB(RFSV,HAR-L)	0.952	**0.982**	0.987	0.952	0.985	1.051	0.941	0.978	**0.986**	0.926	0.970	1.028

Table 4: Test period (2015–), loss relative to HAR, geometric mean over indices. Bold: the model chosen on the validation period for that data set and horizon. Lines below the first block are the improvement candidates.

RFSV against HAR. On RV data the RFSV predictor is 2.4% worse than HAR at one day (better on only 5 of 31 indices; significantly worse on 17), level at five days (1.001) and 0.4% better at twenty days (better on 15 of 31, significant on none). On GK data it is worse at 1 and 5 days on every index and level at 20 days. In QLIKE, RFSV is clearly worse than HAR at 20 days (1.079 on RV), and the best QLIKE at 20 days is the calibrated GARCH (0.983 RV, 0.950 GK), whose level calibration copes best with the 2020 volatility spike. Both RFSV and HAR beat the random walk and GARCH in log-MSE everywhere. For the S&P 500 alone RFSV does beat HAR at one day (ratio 0.967), closer to GJR's SPX result, but this does not generalise across indices.

Data	Comparison (A vs B)	h	log-MSE: A better	log-MSE: worse	QLIKE: A better	QLIKE: worse
RV	RFSV vs HAR	1	2	17	2	6
RV	RFSV vs HAR	5	0	4	1	0
RV	RFSV vs HAR	20	0	0	1	0
RV	HAR-L-RFSV vs HAR	1	29	0	22	0
RV	HAR-L-RFSV vs RFSV	1	30	0	20	0
RV	COMB(RFSV,HAR-L) vs HAR	5	21	0	10	0
RV	COMB(RFSV,HAR-L) vs RFSV	5	19	0	9	0
RV	RFSV-roll250 vs HAR	20	0	1	0	4
RV	RFSV-roll250 vs RFSV	20	0	8	0	14
GK	RFSV vs HAR	1	0	18	0	8
GK	RFSV vs HAR	5	0	4	0	1
GK	RFSV vs HAR	20	1	1	0	0
GK	HAR-L-RFSV vs HAR	1	20	0	20	0
GK	HAR-L-RFSV vs RFSV	1	20	0	19	0
GK	HAR-L-RFSV vs HAR	5	18	0	6	0
GK	HAR-L-RFSV vs RFSV	5	18	0	12	0
GK	COMB(RFSV,HAR-L) vs HAR	20	6	0	0	0
GK	COMB(RFSV,HAR-L) vs RFSV	20	4	0	2	0

Table 5: Diebold–Mariano tests on the test period: number of indices (of 31 RV, 20 GK) where A is significantly better / worse than B at 5%.

6 Improvements

Candidates: (i) RFSV with H, ν from a rolling window of 250, 500 or 1000 days (adaptive H); (ii) HAR-L, HAR with the leverage terms r_t^-, $\overline{r^-}_{t-4:t}$, $\overline{r^-}_{t-21:t}$ (Corsi & Renò, 2012); (iii) HAR-RFSV, HAR with the real-time RFSV forecast as a fifth regressor; (iv) HAR-L-RFSV, both; (v) COMB, $w\,\text{RFSV} + (1-w)\,\text{HAR(-L)}$ with $w \in [0,1]$ re-estimated every day by least squares on past realised forecast errors. One model per data set and horizon is chosen by the lowest validation log-MSE ratio to HAR (validation table in the notebook), then evaluated once on the test period.

Chosen on validation (RV): HAR-L-RFSV at $h = 1$, COMB(RFSV, HAR-L) at $h = 5$, RFSV-roll250 at $h = 20$; (GK): HAR-L-RFSV at $h = 1, 5$, COMB(RFSV, HAR-L) at $h = 20$. On the test period:

	RV, log-MSE / HAR			GK, log-MSE / HAR		
	$h=1$	$h=5$	$h=20$	$h=1$	$h=5$	$h=20$
RFSV (paper)	1.024	1.001	0.996	1.059	1.015	0.996
Selected model	**0.941**	**0.982**	1.008	**0.933**	**0.965**	**0.986**
indices where it beats HAR	31/31	31/31	8/31	20/20	20/20	19/20
indices where it beats RFSV	31/31	30/31	1/31	20/20	20/20	20/20

The improvement beats both benchmarks at 1 and 5 days on both data sets, significantly on most indices (Table 5), and at 20 days on GK (significant against HAR on 6 of 20 indices). At 20 days on RV it fails: the rolling-H RFSV that won the validation period is worse than HAR and than the expanding-window RFSV on the test period. Most of the short-horizon gain comes from the leverage term: HAR-L alone gives 0.954 / 0.987 / 0.992 (RV), and adding the RFSV forecast lowers this to 0.941 / 0.973 / 0.975. (HAR-L-RFSV was also the best test model at 20 days, but it was not the validation choice, so we do not claim it.) In QLIKE the selected models beat HAR at 1 and 5 days (0.946 and 0.985 on RV) but not at 20 days.

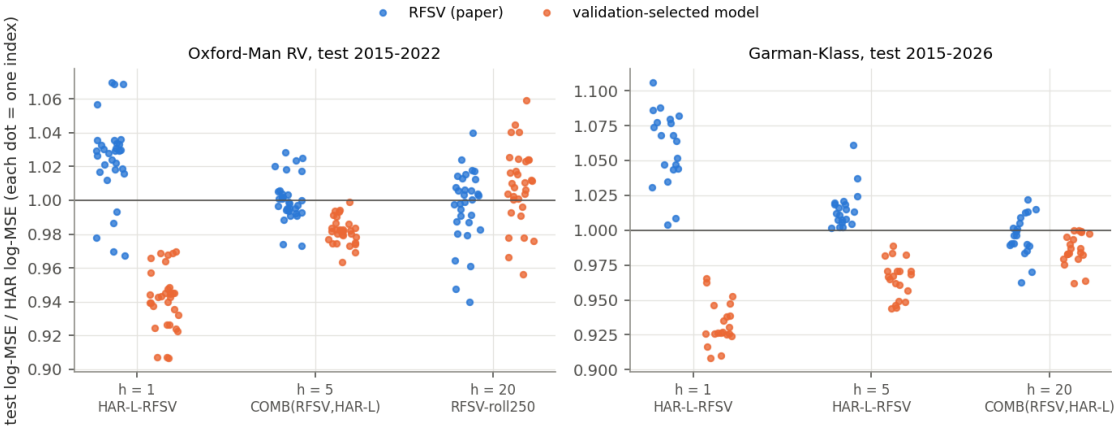


Figure 3: Test-period log-MSE relative to HAR, one dot per index: RFSV (blue) and the validation-selected model (orange).

7 Limitations

- The RV sample ends in February 2022, when the Oxford-Man library stopped; post-2022 results rely on the noisier range-based proxy, whose H is not comparable in level with RV-based H.
- H estimated from daily proxies is biased upwards by daily integration and downwards by noise (Table 3); we did not correct for either.
- The target is point RV h days ahead (as in GJR), not the average over the next h days; HAR is usually specified for the latter, and rankings could differ.

- Variance (QLIKE) forecasts depend heavily on the level correction. The paper's correction (4) uses ν estimated from noisy RV; all models under-predict the mean level in the test period (mean $v/\hat{v}$ above 1, driven by spikes such as March 2020).
- Indices are not independent (cross-market correlation), so the DM counts are not independent tests. No Model Confidence Set and no multiple-testing correction across the 8 candidates (the validation step is the only guard against selection bias).
- Yahoo index OHLC is not an audited source; we cleaned obvious errors only.

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